

## EDUCATION

### Ph.D. in Civil and Environmental Engineering

The University of Iowa CGPA: 3.89/4.00

08/2024 - Present

IA/USA

### M.Sc. in Civil and Environmental Engineering

The University of Iowa CGPA: 3.89/4.00

08/2024 - 05/2026

IA/USA

### B.S. in Civil Engineering

Middle East Technical University, METU

2017 - 2022

Ankara/TURKIYE

## RESEARCH EXPERIENCE

Graduate Researcher, Prof. Dr. Ibrahim Demir Research Group, IIHR—University of Iowa (2022 - Present) [🔗](#)

- Comprehensive Analysis of Riverine Flooding Impact on Bridges: Iowa Case Study (*Published*)
- Comprehensive Flood Impact Assessment for Iowa Bridge Infrastructure Using Integrated AHP and Fuzzy AHP Analysis (*Manuscript in Review*)
- Enhancing the Resilience of Wind Energy Infrastructure in Iowa Flood Risk Assessment and Site Suitability Analysis for Critical Infrastructure Protection (*Published*)
- IBIS: A Community-Oriented Framework for Flood-Induced Bridge Vulnerability Assessment and Prioritization for Iowa (*Manuscript in Review*)
- Flood Damage & Vulnerability Assessment of Electrical Substations in Iowa (*Manuscript in Preparation*)
- Comprehensive Spatial Analysis for Farm-Related Accidents: U.S. Midwest Region Case Study (*Manuscript in Preparation*)
- Spatial Assessment of Urban Flood Vulnerability in Polk County, Iowa Using a Social-Ecological-Technological Framework (*Manuscript In Review*)

Undergraduate Researcher, Prof. Dr. Ahmet C. Yalciner Design Project Group, METU (2022)

- Marina design for Santorini Island, Greece (volcanic Mediterranean setting), involving coastal dynamics analysis, breakwater layout, and marine foundation design under wave-structure interaction. Further information: <https://drive.google.com/file/d/1vIOSiOCxHBB1dDWfdayGos973UykQlfo/view?usp=sharing>

## INTERNSHIP

### Academic Project Engineering Intern

Middle East Technical University, Department of Civil Engineering

2021

Ankara/TURKIYE

Gained practical experience with Bentley OpenRoads, Microsoft Excel, SAP2000, Primavera P6, and Rocscience Slide2. Further Information: <https://drive.google.com/file/d/14ySulCdraWL96NnFiogbt-q4EKoTbY2C/view?usp=sharing>

### Site Engineering Intern

Folkart Yapi Construction

2020

Izmir/TURKIYE

Construction of the Folkart Vega Project, comprising 3 blocks, 812 residences, and 42 commercial units in Izmir, Türkiye. Further information: <https://drive.google.com/file/d/1qLl6ixfl4w7S5jbue4qEoqehQegGajp1/view?usp=sharing>

## SOFTWARE SKILLS

QGIS (proficient)

ArcGIS (proficient)

PHP (proficient)

JS (proficient)

HTML/CSS (proficient)

SQL (proficient)

MATLAB (proficient)

MS Office & Project (proficient)

AutoCAD (proficient)

Revit (good)

HEC-RAS (good)

EPANET (good)

Python (good)

SAP2000 (good)

ENVI (good)

BENTLEY OpenRoads (good)

Slide2 (good)

## PUBLICATIONS

Duran, E., Cikmaz, A. B., Mount, J., & Demir, I. (2026). Spatial Assessment of Urban Flood Vulnerability in Polk County, Iowa Using a Social-Ecological-Technological Framework. *Natural Hazards* (In Review)

<https://doi.org/10.31223/X55771>

Duran, E., & Demir, I. (2026). Enhancing the Resilience of Wind Energy Infrastructure in Iowa: Flood Risk Assessment and Site Suitability Analysis for Critical Infrastructure Protection. *International Journal of Disaster Risk Reduction*, 106003

<https://doi.org/10.1016/j.ijdrr.2026.106003>

Duran, E., Mount, J., & Demir, I. (2025). IBIS: A Community-Oriented Framework for Flood-Induced Bridge Vulnerability Assessment and Prioritization for Iowa. *Journal of Hydroinformatics* (In review)

<https://doi.org/10.31223/X5CT9H>

Duran, E., Mount, J., & Demir, I. (2025). Comprehensive Flood Impact Assessment for Iowa Bridge Infrastructure Using Integrated AHP and Fuzzy AHP Analysis. *Resilient Cities and Structures* (In review)

<https://doi.org/10.31223/X5VD9Q>

Duran, E., Alabbad, Y., Mount, J., Yildirim, E., & Demir, I. (2025). Comprehensive Analysis of Riverine Flood Impact on Bridge and Transportation Network: Iowa Case Study. *International Journal of River Basin Management*, 1–14.

<https://doi.org/10.1080/15715124.2025.2457546>

## CONFERENCES

Duran, E., Mount, J., & Demir, I. (2025). Flood Impact Analysis for Iowa Bridge Infrastructure Using Integrated AHP and Fuzzy AHP Approaches. In 6th IAHR Young Professionals Congress (pp. 148–149). ISBN: 978-90-834302-5-6.

## EXTRACURRICULAR ACTIVITIES

Player, METU Soccer & Futsal Teams—Achieved 3rd place in the Turkish University Teams Championship. (2017-2022)

Organization Committee Member, METU International Association of Civil Engineering Students (2017-2019)